Lightning fast computing

Siddharth Parwatay
siddharth.parwatay@thinkdigit.com

Technology on the nigh horizon will make computers faster than ever before. Read on and get a glimpse into a blazing future

From the bygone era of early computing, when clunky behemoths of first generation computers were considered fast, to the pocketable miracles of today; we have come a long way. Computers have grown to be faster, smaller, and more energy efficient almost exponentially, and keeping in tune with Moore's Law. But as chip sizes go smaller, in an effort to pack in more and more circuitry to speed things up, fundamental physical restrictions come in the way. The chips start leaking signals and won't hold digital information anymore. So then, are we at the frontiers of what can be done? Not at all. We've just barely begun to scratch the surface to unlock the secrets that computing can unfold. Bottlenecks will be a thing of the past, internal data speeds will increase beyond our wildest imagination, computers will be able to keep up with our thoughts, and the very definition of computing will change. Here's a look at some of the technologies on the drawing board which inventors, scientists and futurists alike are excited about that will make computing faster than ever before.

DNA computers

A fusion of technology and bionics has for long been the holy grail for science. One of the explorations in this field is the possibility of using DNA to perform complex calculations. If you look at conventional computers, they're at their heart machines, that perform calculations; even when they're playing your favourite MP3 track. These calculations are performed via electronic impulses that run across silicon circuitry. Again, the limitations of silicon prompted researchers to look at alternate materials. And they couldn't have come up with a more absurd one – DNA – essentially the building blocks of life. When you give it further thought, it's but natural. After all, genes do store a hell of a lot of data and can perform millions of calculations every minute throughout an organism's life. In theory we're walking around with billions of supercomputers inside us. In DNA computers, DNA molecules can be used to perform calculations using enzymes. When they undergo reactions at a molecular level, calculations and simple operations are a byproduct. To demonstrate this in 2002 scientists from the Weizmann Institute of Science in Rehovot, Israel created a computing machine based on this principle to solve simple mathematical problems. The composition of the DNA can be manipulated to control the type of output. The speed advantage here is that unlike conventional computers that operate linearly, billions of molecules will attack the problem parallelly. The problem that such computers could solve are those that involve infinitely large number of possibilities such as weather prediction for instance. Problems that conventional computers might take years to perform would be solved in a couple of hours. Wonder if they'll play a 3D blu-ray though?

Quantum Computers

Just like DNA-based computers the idea of Quantum Computers has been around for some time now, but with each passing year, it comes closer to reality. Quantum mechanics is all about entering the subatomic world of particles such as neutrons and electrons. These computers will make use of